tec-speed 30.0 VT-6702

UL Approval: E214381 Version:20/11/2024

PROCESS GUIDE

Precautions in Handling

Storage Condition & Shelf Life

		Laminate
Storage Condition	Temperature	Room
	Relative Humidity	/
Shelf Life		24 Months(airproof)

- Take care in handling thin core laminates as they are easily damaged.
- The grain direction is indicated on the label with an arrow.
- Laminates exceeding 24 months can still be used as long as the copper foil has no severe oxidation or stain.

Inner Layer Process

- Tooling & pinning holes should be drilled or routed. Punching is forbidden.
- Lay the panel on a flat clean surface. The solvent clean should be followed with a clean water rinse.
- Remove copper oxides using commercially available cleaners and micro etchants.
- Abrasive cleaning (e.g., pumice powder or conveyorized brush scrubbing) is not recommended.
- Mechanical surface preparation tends to cause microscopic scratches in copper. These scratches become stress risers that can lead to thermal stress cracks in some environmental conditions. While strongly recommending against abrasive cleaning, we recognize that in some circumstances that abrasive cleaning is necessary. When abrasive cleaning is required, it should be performed with minimal pressure and knowledge of the risks involved. Brush scrubbing may cause material distortion.
- After oxide, it is preferred to press cores within 12 hours. Any cores staged more than 24 hours should be re-treated.

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Laminates Availability

Laminate	Core Thickness (inch)	Dk @ 10GHz	Df @ 10GHz
VT-6702	0.010	2.94±0.04	0.0011
	0.020	2.94±0.04	0.0011
	0.030	2.94±0.04	0.0011
	0.060	2.94±0.04	0.0011

Remarks:

- Copper foil: Hoz~2oz, HTE, RTF and HVLP are all available based upon request.
- More types may be available upon request.
- Metal base and Resistive Cu are available upon request.

Typical Drilling Parameters

Drill Bit Diameter		Spindle (KRPM)	Infeed (IPM)	Retract (IPM)	Hit Counts
0.50mm	0.0197in	48	95	500	400
0.65mm	0.0256in	37	74	500	400
1.00mm	0.0394in	24	48	500	400
1.25mm	0.0492in	20	40	500	400
1.59mm	0.0625in	20	40	500	400
2.00mm	0.0797in	20	40	500	400
3.00mm	0.1181in	20	40	500	400

- Chip load: 0.002”~0.004” / revolution.
- Surface speed: 200~300 SFM.
- Excessive wear of carbide drill bits. Carbide or Diamond coated drill bits are recommended.
- LE Aluminum sheet and white phenolic entry board are preferred, please consult your local Ventec representative for options.

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PROCESS GUIDE

Desmear Process

Plasma treatment is necessary process, and material should be baked more than 1 hour at 120°C before plasma treatment. Meanwhile, material cannot be suffered any high pressure washing and scrubbing before metallization.

Plasma: Half time of FR-4 conditions is recommended.

Pressure (mT)	Temperature (°C)	Gas mixture	Power (W)	Duration (min)
250	90	90%H_2 , 10% N_2	4500	5
250	100	10%CF_4 , 90% N_2	4000	10
250	100	100% O_2	4000	5
250	100	100% N_2	4200	30

Permanganate Desmear: Not recommended.

Plating and solder mask

- VT-6702 laminates have been engineered to be highly resistant to chemical infiltration and a robust hole wall preparation process is necessary for a successful deposition plating process. Following hole wall preparation, VT-6702 will accept standard electroless copper or direct deposit metallization plating process.
- When copper surface preparation is required, chemical cleaning processes are preferred.Mechanical scrubbing should be avoided due to possible mechanical damage. Otherwise, standard processing should be used.
- No special treatment is required if the surface has not been mechanically scrubbed. In rare cases, where adhesion is poor, a plasma treatment may be used to activate the exposed PTFE.

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Routing

Drill Bit Diameter (mm)	Spindle (KRPM)	Infeed (IPM)	Retract (IPM)
0.8	50	10	500
1.0	40	12	500
1.2	34	12	500
1.6	25	18	700
2.0	20	27.2	700
2.4	20	35	700
3.0	20	42	700
3.2	20	42	700

- Excessive wear of carbide drill bits and Diamond coated drill bits are recommended.
- Rigid phenolic entry and a rigid backer should be used. In some cases, adding paper (white paper or Kraft paper) between the phenolic and the part allows better conformance to surface topography and may reduce burring.
- Materials will generally provide a better edge quality due to none glass fabric.

Packaging and baking recommendation

- It is recommended to bake the board before packaging at 125°C/4~8h to avoid moisture causing a decrease in heat resistance.
- If the PCBs needs to be stored for a long time before use, it is recommended to use aluminum foil vacuum packaging.
- If exceed 3 months after packaging, It is best to bake the PCBs at 125°C/4~6h before assembly before use.